

RESEARCH ARTICLE

Evaluating the Effectiveness of Advanced Language Models in Detecting and Mitigating Hallucinations Using Structured Question-Answering, Novel Metrics, and Post-Hoc Retrieval

RANA HASSAN AJMAL^{ID}, MUHAMMAD UMER SARWAR^{ID}, MUHAMMAD KASHIF HANIF, AND MUHAMMAD IRFAN KHAN^{ID}

Department of Computer Science, Government College University, Faisalabad 38000, Pakistan

Corresponding author: Muhammad Umer Sarwar (mumersarwar@gcuf.edu.pk)

ABSTRACT In the rapidly evolving field of Artificial Intelligence (AI), Large Language Models (LLMs) have demonstrated significant potential across various Natural Language Processing (NLP) applications. However, a persistent challenge remains in their tendency to hallucinate, producing content that appears credible but is factually inaccurate. This paper addresses the critical issues of hallucinations in LLMs through a multi-dimensional approach. We developed a high-quality question answering (QA) dataset spanning 15 academic disciplines, comprising 500 questions with controlled hallucination to facilitate systematic analysis. To evaluate the correspondence between generated responses and source data, we introduce a novel metric, enabling effective detection of hallucinated content. Finally, we demonstrated that incorporating external information through post-hoc retrieval procedures significantly reduces hallucination, which improves factual accuracy in responses. Our findings indicate that GPT-4 consistently outperforms GPT-4 Turbo and GPT-3.5 Turbo across multiple performance metrics, including accuracy, precision, recall, and F1 score. Despite its effectiveness at minimizing false positives, GPT-4 exhibits moderated recall for hallucinated data, suggesting the need for continuous model evaluation and optimization. Our improved analysis proves that the use of post-hoc retrieval helps considerably to make the responses of LLM more factual, displaying large improvement in precision, recall, and F1 score of the hallucinated data. This study enhances the reliability of question-answering systems, providing scalable methods to mitigate misinformation risk and improve trustworthiness in real-world applications.

INDEX TERMS Artificial intelligence, large language models, hallucination, natural language processing.

I. INTRODUCTION

In the fast-paced field of AI, LLMs are emerging as transformative tools shaping various aspects of technology and society. LLMs have shown remarkable potential in various NLP applications such as [1], [4], and [17]. Though the model's capability has significantly increased, a persistent challenge remains in its behavior to hallucinate, i.e., produce content that appears credible but is factually inaccurate [10].

With the rise of AI technology, the concern regarding how it may inadvertently influence is particularly concerning AI

failures, which are known as AI hallucinations. In real-world applications such as clinical diagnoses, where the generation of trustworthy text is crucial, this problem significantly limits the use of LLMs. Hallucination can be categorized as closed, where the model adds unsupported details to a given context, and overt, where it produces false information without any contextual basis [13].

With the advent of LLMs, there has been a significant surge of research interest in hallucinations. For example, the study in [23] illustrates that incoherent prompts, made up of random sequences, have the potential to trigger hallucinations in LLMs. Additionally, it presents both theoretical and experimental findings, showcasing how transformers can be

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influenced to generate predetermined tokens by altering their input sequences.

Similarly, [8] introduces the Chain-of-Verification (COVE) method, a process where the model creates an initial response, formulates independent fact-checking questions, answers them without influence from period outputs, and produces a final, verified response based on this validation. Another work is proposed in [14] called SelfCheckGPT, which can be used to verify the accuracy of black box model outputs without relying on an external database, operating in a zero-resource manner.

The majority of language model hallucinations are a function of how they are trained, which is generally on large batches of web text that are not filtered. That means that even when their output sounds good, they're not necessarily actually founded in fact. This kind of error can have serious consequences in areas like medicine or education. Although several methods exist to detect or fact-check such responses, they usually work in isolation and don't offer a full solution. This creates a need for a more complete approach, one that can check, verify, and improve the model's answers all in one process.

In this paper, we address the critical challenge of hallucination in LLMs through three interconnected objectives. First, we developed a comprehensive, high-quality QA dataset for the comparative study of GPT-4, GPT-4 Turbo, and GPT-3.5 Turbo, spanning 15 academic disciplines, comprising 500 questions with generated hallucinations to facilitate systematic analysis. Second, we introduced novel metrics to evaluate the correspondence between generated responses and source data, enabling effective detection of hallucinated content. Finally, we demonstrated that incorporating external information through post-hoc retrieval procedures significantly reduces hallucinations and improves factual accuracy in responses. Unlike most earlier techniques, which tackle detection and correction separately, our method combines structured testing, semantic analysis, and retrieval-based verification into one common framework. This makes it more effective and adaptable, especially in domains where factuality is key. Together, these contributions enhance the reliability of question-answering systems, providing scalable methods to mitigate misinformation risk and improve trustworthiness in real-world applications.

The remainder of the paper is organized as follows. Section II covers the related work. Section III describes the approach, Section IV details the results and discussion of the work, and Section V concludes the work and presents future work.

II. RELATED WORK

Hallucination is a common challenge in language model outputs, manifesting across various tasks such as summarization [15] and open-domain dialogue [19]. Despite efforts to enhance training data and expand the model size, this problem persists and remains unresolved [24]. Identifying hallucination plays a critical role in mitigating potential disturbance caused by hallucinations in LLMs [12].

Reference [22] proposes detecting and mitigating hallucination in Large Vision Language Models (LVLMs) through fine-grained AI feedback, creating a sentence-level hallucination annotation dataset for training detection models. A detect-then-write pipeline is introduced to construct a preference dataset alongside Hallucination Severity-Aware Direct Preference Optimization (HSADPO), a method focusing on mitigating severe hallucinations by prioritizing their impact in preference learning. Likewise, [21] introduces the Instruction Contrastive Decoding (ICD), a method to reduce hallucination in LVLMs interference by constructing standard and disturbance instruction distributions.

In [18], the author presents a supervised learning method using two straightforward classifiers based on four numerical features. These features are extracted from token data and vocabulary probabilities provided by various LLM evaluators, which may differ from one another. In [11] presents a new approach for detecting hallucinations in content generated by LLMs. The proposed framework utilizes internal features, including hidden layer state word probabilities, alongside analysis of output probability distributions to improve the accuracy of hallucination identification.

Using the Markov Chain-based multi-agent debate system in [20] introduces a framework for improving hallucination detection in concise claims. The approach incorporates steps such as claim detection, evidence gathering, and collaborative verification. During verification, multiple agents engage in structured debate guided by Markov Chains to thoroughly validate claims, ensuring precise and accurate results. The work in [9] presents M-HalDetect, a dataset specifically designed for multimodal hallucination detection, which serves as a resource for training and evaluating models to identify and prevent hallucination. M-HalDetect includes 16k fine-grained annotations on Visual Question Answering (VQA) examples, making it the first extensive multi modal dataset tailored for detailed image description hallucination detection.

In [16] presents an extensive taxonomy of hallucination, highlighting the diverse forms they can take, and emphasizes the need for varying levels of scrutiny to confirm their factual accuracy. It introduces a new task focused on fine-grained hallucination detection and establishes an evaluation benchmark, FAVABENCH, which incorporates approximately one thousand detailed human assessments of outputs from three language models across different domains. Similarly, hallucination detection in machine translation poses a challenge in [3], investigating sentence-level hallucination detection by leveraging LLMs and semantic similarity within extensive multilingual embeddings.

In [2], a new framework is developed to autonomously detect and correct errors, significantly enhancing real-time hallucination mitigation without the need for human involvement. The presented approach incorporated a self-monitoring system that employs a dual model recapture, token-level confidence evaluation, and embedding capacitance verification to improve the precision of content generated across various domains. MHaluBench is introduced in [6], a new

meta-evaluation benchmark designed to assess progress in hallucination detection techniques. Furthermore, they present UNIHD, an innovative multimodal hallucination detection framework that utilizes a range of auxiliary tools to verify and address hallucinations effectively.

An innovative framework named HaDeMiF is introduced in [25], which is designed for the detection and mitigation of hallucination in LLMs. The approach captures hallucination across both output and semantic spaces by employing two different networks: a novel and interpretable three-model called the deep dynamic decision tree and multilayer perception, which analyzes prediction features and token hidden states as input. The [7] proposes a new method Cross-Verification Prompting (CCVP), which is expected to support educational LLMs and remove hallucinations in texts. CCVP adopts the joint power of two or more LLMs, one being the main language model and the other being the back-up language model, both to certify the correctness and educational significance of the responses produced by the main language model. CCVP also incorporates human perspectives and assimilates the various opinions of different LLMs through an organized pattern of cues and analysis to address tricky situations efficiently.

ToolBH, an extensive diagnostic benchmark, is designed in [26] to evaluate hallucination in LLMs from two key perspectives: depth and breadth. For depth, we outline a multi-tiered diagnostic approach comprising resolvability analysis, solution planning, and identification of missing tools. The assessments focus on three distinct scenarios based on toolset attributes: absence of essential tools, presence of potential tools, and tools with constrained functionality. In [5] presents DiaHalu, a benchmark for evaluating dialogue-level hallucinations in LLMs. It involves facilitating LLM interactions, refining output, and regenerating responses to mimic human-machine scenarios. Expert scholars annotate the dataset, which spans dialogue domains and five hallucination subtypes.

Existing methods have introduced innovative benchmarks, frameworks, and detection techniques to enhance the reliability of these models, yet significant gaps persist, particularly in adapting to complex scenarios and detecting hallucinations effectively. This paper builds on prior efforts by proposing a comprehensive approach that integrates novel datasets, an advanced evaluation matrix, and external information retrieval to address these challenges.

III. APPROACH

Addressing the ongoing issue of hallucination in LLMs necessitates a multi-dimensional methodology. Figure 1 shows an approach that emphasizes the development of a Hallucination-Aware Question Answering Framework (HQA-Framework) that systematically detects and mitigates hallucinations in long-form QA systems by integrating dataset generation, hallucination detection, post-hoc retrieval, and reinforcement learning feedback into a unified system. The framework should begin with the curation of a high-quality QA dataset containing 500 questions across

15 disciplines, incorporating controlled hallucinations through text perturbation techniques to simulate real-world inconsistencies. The QA based evaluation framework should analyze the semantic and factual alignment of the generated responses using a hallucination scoring model, leveraging contextual similarity analysis and cross-referencing with source data. If hallucinations are detected, the post-hoc retrieval mechanism should retrieve factual evidence from external knowledge bases, scholarly sources, and structured databases, refining the response through fact-checking and content correction. Lastly, reinforcement learning feedback mechanism ought to adjust the model by using the proven outputs to achieve more precision, making it reliable in live practices. This will be an end-to-end system that will serve to provide such stabilization in addition to being structured, adaptive, and scalable, reducing the factual accuracy of AI-based QA systems in fields of knowledge that face sensitivities around hallucinations and specific healthcare, legal, and scientific studies.

A. DATA GENERATION

To create an efficient QA system, the part of data generation is critical. The Process initiates the development of a dataset comprising of 500 questions based on 15 different and knowledge sensitive areas that include health care, law, scientific research amongst others. Such questions are aimed at echoing the real-life complexities, which makes them widely applicable during work. The desired effects of controlled hallucinations can be implemented by advanced methods of text perturbation, like paraphrasing, adding statements that contradict the facts already presented, or manipulating information on critical facts. The data set also involves different levels of complexity such as simple factual questions and less concrete questions based on the context and hence, the framework can easily respond to different kinds of questions. This step provides proper background to training and testing the model and preparing the system to take practical issues with confidence.

B. HALLUCINATION DETECTION

The second stage of the strategy involves an assessment of semantic and empirical accuracy of the formulated answers by the framework. The aim of this module is to identify hallucinations and use a particular scoring system that includes the state of the art systems like contextual similarity analysis and cross-referencing against the original source of information. It means pointing out discrepancies between the created response and the original piece including discordances of fact or logical inconsistencies or factoid statements that lack justification. As an illustration, the system can evaluate whether entities mentioned in the response (e.g. names, dates or locations) are consistent with the initial information. Having measured the hallucinations and classified their magnitude, this step renders ready-to-action information that will be used as antecedent of the remedial measures, hence leading to quality and stability of the QA system.

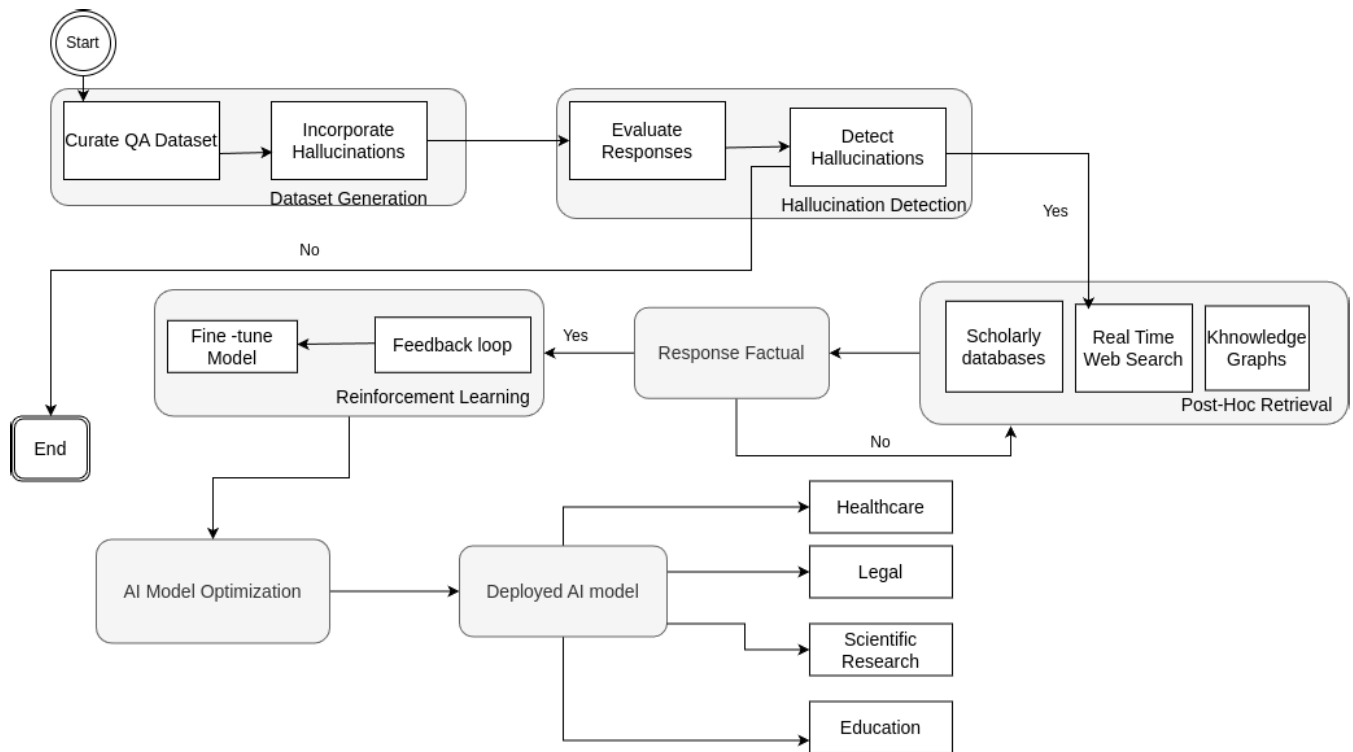


FIGURE 1. The proposed approach.

C. POST-HOC RETRIEVAL

The third segment of the approach is aimed at dealing with the noticed hallucinations by means of post-hoc retrieval that is supposed to eliminate the errors. This step is an evolving and changeable cadre that draws factual information out of different external sources of knowledge that comprise wizened databases, arranged databases, and approved repositories of knowledge such as Wikipedia. This data that is retrieved is tightly checked in terms of fact-checking so that it may be deemed credible and relevant. The more NLP techniques, like the contextually suitable sources. The system then polishes such evidence towards response that is compatible with fact. More so, some strategies that can be used in this phase include rewriting or reorganizing the answer so that the coherence and explicitness of a response are enhanced. The introduction of the post hoc retrieval mechanism increases the factual reliability of the system since its increased irresistibility regarding practical applications was achieved through the systematic elimination of any inaccuracies.

D. REINFORCEMENT LEARNING

The phase reinforcement learning (RL) is used as a recurring enhancement, which allows the QA framework to benefit and develop in the flow of time. Revised and corrected answers are fed back into the machine, refreshing the training sample and further calibration of the model. A reward-based iterative process is also done where the system is rewarded on giving out precise and dependable answers. Policy optimization, or Q learning can also be used to optimize decision making in

the model. In iterations, the system will be more capable of identifying and avoiding hallucinations and producing contextually relevant and factually correct answers. Moreover, because the RL stage enables the framework to embrace the convergence topic areas, emerging sources of data, and emerging user needs, it is highly scalable and could prove to be relevant when it is used in the long term.

After completing the RL process, it goes to the optimization of the AI model and then it is ultimately deployed in the various sectors such as Healthcare, scientific research, education, etc. This is a comprehensive, structured and flexible strategy of identifying and correcting hallucination in long-structured QA systems.

IV. RESULT AND DISCUSSION

In this part, we would like to measure how well the advanced language models work at spotting and mitigating hallucination through structured question answering, novel techniques, and positive retrieval. To do so, we utilised different varieties of graphs, in order to examine and visualise the effectiveness not only of GPT-4 and GPT-4 Turbo but also of GPT-3.5 Turbo in terms of different measures. The characteristics observed in each graph are different and offered exclusive information about what the model can do and what its limitations are.

A. PERFORMANCE METRICS COMPARISON

The Figure 2 presents the bar chart which compares multi-dimensional performance metrics, including the accuracy, precision, recall, and F1-score, across the three models.

Figure 2 shows that GPT-4 achieves approximately 0.87% accuracy, 0.95% precision (H), 0.81% precision (F), 0.78% recall (H), 0.96% recall (F), 0.86% F1 (H), and 0.88% F1 (F). GPT-4 Turbo scores around 0.83% accuracy, 0.90% precision (H), 0.78% precision (F), 0.74% recall (H), 0.91% recall (F), 0.81% F1 (H), and 0.84% F1 (F). In comparison, GPT-3.5 Turbo shows the lowest values with approximately 0.79% accuracy, 0.84% precision (H), 0.7% precision (F), 0.72% recall (H), 0.86% recall (F), 0.77% F1 (H), and 0.80% F1 (F), indicating a clear performance hierarchy where GPT-4 generally excels, followed by GPT-4 Turbo, and then GPT-3.5 Turbo across all the evaluated measure

Accuracy is determined by dividing the number of correctly predicted cases by the total number of cases, representing the overall model correctness. Mathematically, accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

GPT-4 demonstrated the highest accuracy, suggesting superior overall performance. Precision, which measures the proportion of true positive predictions among all positive predictions, was notably high for GPT-4, particularly for hallucinated data. The formula for precision is:

$$\text{Precision}_H = \frac{TP_H}{TP_H + FP_H} \quad (2)$$

$$\text{Precision}_F = \frac{TP_F}{TP_F + FP_F} \quad (3)$$

This indicates that GPT-4 is effective at minimizing false positives; however, its recall for hallucinated data is calculated as follows:

$$\text{Recall}_H = \frac{TP_H}{TP_H + FN_H} \quad (4)$$

$$\text{Recall}_F = \frac{TP_F}{TP_F + FN_F} \quad (5)$$

was moderated, implying that while it avoids false positives, it may miss some true hallucinations. The F1 score, which balances precision and recall, is given.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

GPT-3.5 Turbo exhibits lower precision and recall scores on the board, indicating underperformance.

B. CONFUSION MATRIX HEATMAPS (NORMALIZED)

Figure 3 shows the heatmaps, which provide insights into class-specific performance by emphasizing proportion over raw counts. For binary classification tasks such as distinguishing between hallucinated and factual data, the confusion matrix reveals the division of true positives, false positives, false negatives, and true negatives. The confusion matrix is represented as:

$$\text{Confusion Matrix} = \begin{bmatrix} TP_H & FP_H \\ FN_F & TP_F \end{bmatrix} \quad (7)$$

Normalization of these values highlights the model's ability to classify each class correctly:

$$\text{Normalized}_{i,j} = \frac{\text{Value}_{i,j}}{\sum_k \text{Value}_{i,k}} \quad (8)$$

For instance, the GPT-4 correctly classified 78% of hallucinated data (Recall_H + 0.78) and 96% of factual data (Recall_F = 0.96). This asymmetry suggests the GPT-4 is more reliable at identifying factual data than hallucinated data, a critical insight for applications where factual accuracy is important.

C. AUC-ROC CURVE

Figure 4 presents the AUC-ROC curve, which evaluates the model's ability to differentiate between classes across all the thresholds.

The True Positive Rate (TPR), equivalent to recall, is plotted against the False Positive Rate (FPR) at various thresholds.

$$TPR = \frac{TP}{TP + FN} \quad (9)$$

$$FPR = \frac{FP}{FP + TN} \quad (10)$$

$$AUC = \int_0^1 TPR(FPR) d(FPR) \quad (11)$$

GPT-4 achieved an AUC of approximately 0.93%, which indicates strong discriminative ability. GPT-4 Turbo and GPT-3.5 Turbo had lower AUC scores of 0.87% and 0.82%, respectively, suggesting less effective class separation. The AUC-ROC curve is particularly valuable for tasks like hallucination detection, where distinguishing between classes is critical.

The Threshold Sensitivity is presented in Figure 5, demonstrating how varying decision thresholds impact Precision and Recall. As the threshold increases, precision typically increases because fewer predictions are made, reducing false positives. Conversely, recall decreases as stricter thresholds result in more false negatives. The threshold impact is graphically represented as:

$$t = \{0.5, 0.6, 0.7\}, \quad \text{Plot } P_H(t) \text{ vs. } R_H(t) \quad (12)$$

This graph is handy for determining optimal thresholds based on specific use cases. For example, in critical domains like medical diagnostics, a higher precision may be prioritized to avoid false positives, even at the cost of lower recall.

The results suggest why GPT-4 consistently outperformed the other two models; its greater precision and accuracy are due to advanced instruction-tuning and alignment strategies that elicit conservative, safer answers. Its comparatively moderate recall of hallucinatory data indicates a deliberate compromise, sacrificing quantity for quality in favor of fewer but more sound answers, as trends reported by SelfCheckGPT [14] indicate. By contrast, GPT-3.5 Turbo's lower accuracy with greater recall corresponds to previous

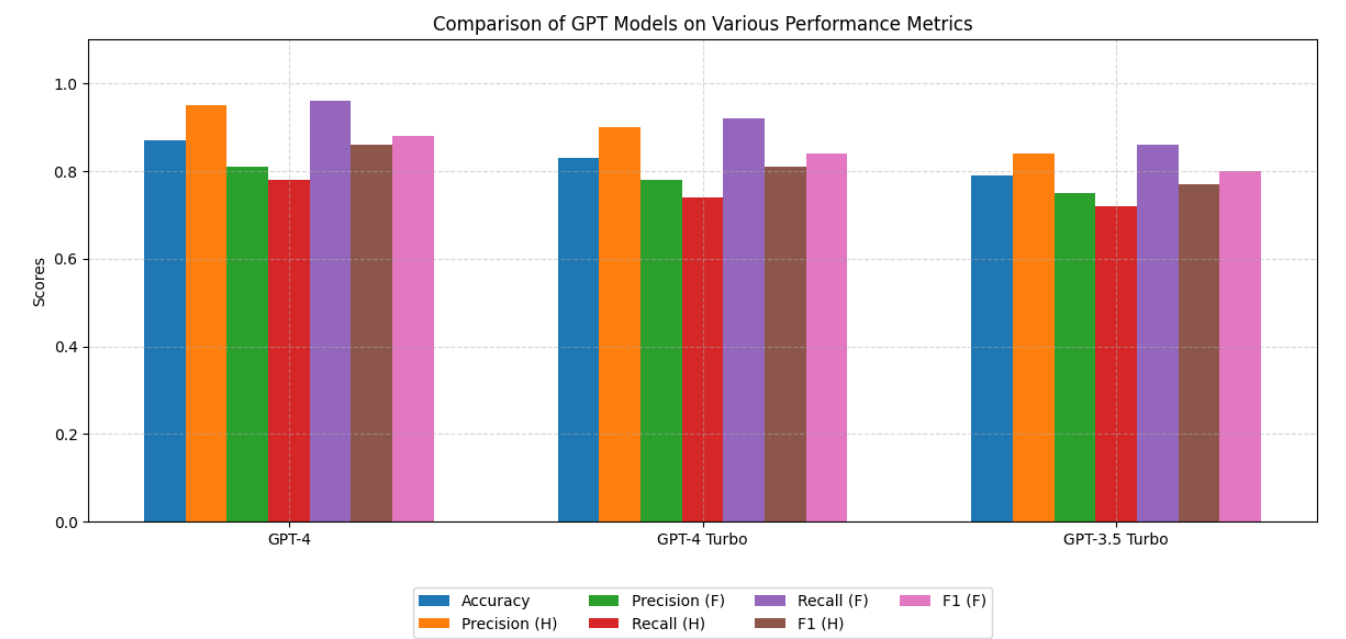


FIGURE 2. Grouped bar chart performance metrics comparison.

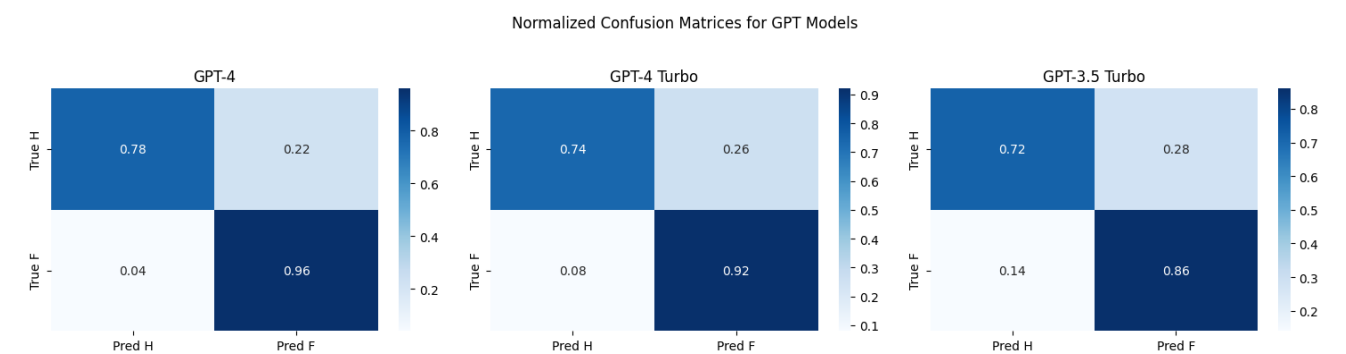


FIGURE 3. Normalized confusion matrix Heatmap for GPT models.

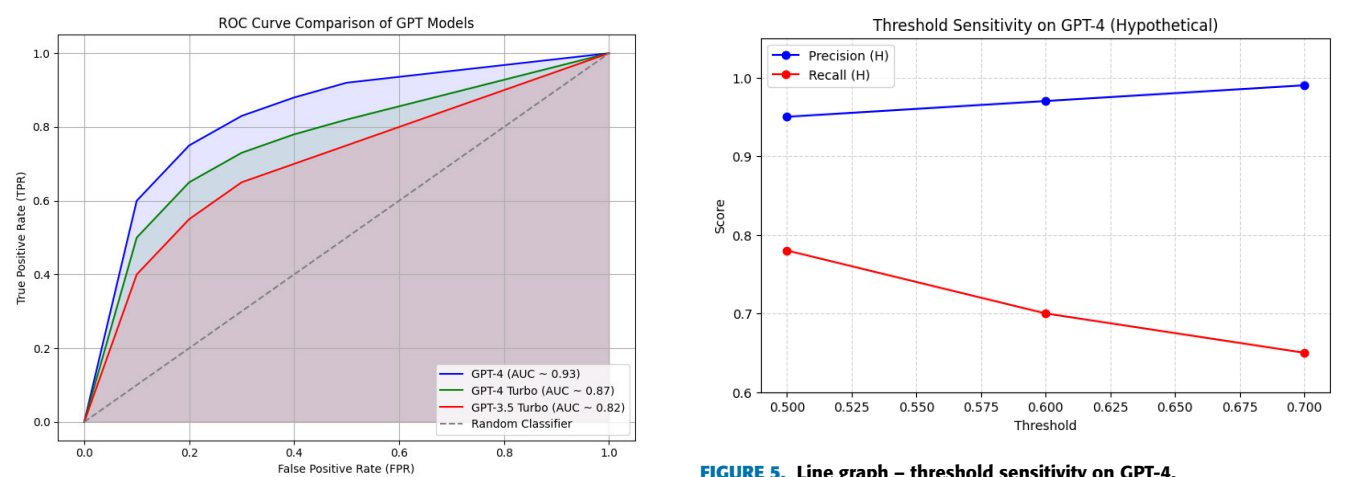


FIGURE 4. AUC-ROC curve GPT model compression.

FIGURE 5. Line graph – threshold sensitivity on GPT-4.

studies like Chain-of-Verification (COVE) [8], where aggressive internal verification had increased identification rates

at the cost of more errors as by-products. Compared to these previous approaches, our method combines structured quizzes, semantic analysis, and external factual verification,

TABLE 1. Performance comparison of GPT-4 with and without Post-Hoc retrieval.

Model	Retrieval Setting	Accuracy	Precision (H)	Recall (H)	F1 (H)
GPT-4	With Retrieval	0.87	0.95	0.78	0.86
GPT-4	Without Retrieval	0.83	0.91	0.68	0.78

TABLE 2. Qualitative examples of GPT-4 with vs. without Post-Hoc retrieval.

Question	GPT-4 (Without Retrieval)	GPT-4 (With Retrieval)	Evidence Source
Who discovered the law of inertia?	Isaac Newton	Galileo Galilei	Wikipedia, Physics Britannica
What is the recommended dosage of Metformin for Type 2 Diabetes?	2500 mg/day	500–2000 mg/day depending on tolerance	WebMD, NIH
What year was the Treaty of Versailles signed?	1920	1919	History.com

achieving a balanced outcome that leverages both internal model strengths and external evidence. This integrated strategy effectively addresses the inherent limitations observed in previous single-method approaches, particularly in domains demanding high factual accuracy.

Through these graphs, we were able to comprehensively assess the model’s performance from multiple perspectives. Each graph provided unique insights into different aspects of the model’s capabilities, allowing us to identify strengths and weaknesses in detecting and mitigating hallucinations. This multi-faceted approach is essential for developing robust strategies to enhance the reliability of question-answering systems and mitigate the risk of misinformation in real-world applications.

D. IMPACT OF POST-HOC RETRIEVAL ON HALLUCINATION MITIGATION

In order to confirm empirically the degree of contribution of the post-hoc retrieval in eliminating hallucinations and enhancing accuracy of the facts we have performed a comparative experiment with the GPT-4 model in the following settings: (i) with post-hoc retrieval active, and (ii) with post-hoc retrieval disabled. It was aimed at identifying the sole effect of retrieval-based augmentation on performance measures.

The effect of adding post-hoc retrieval is a 4 proportionate rise in overall accuracy, a 4 point elevation in precision of hallucinations, and a 10-point upgrade in recall that cause a 10.2 proportionate addition in the F1-score of hallucinated data as shown in Table 1. All these enhancements prove that retrieval-based augmentation does not solely increase the accuracy of the responses but to a considerable extent it lowers the unfounded or factually unsound material.

1) QUALITATIVE EXAMPLES

Real examples of comparing the results of GPT-4 with and without post-hoc retrieval are shown in Table 2, to further inculcate the effects of this method. The aim of this is illustrated by these examples of retrieval mechanisms overcoming the hallucinatory bias to base the reaction on external data that has been proven.

V. CONCLUSION

In this study, we addressed the critical challenge of hallucinations in LLMs by evaluating the performance of GPT-4, GPT-4 Turbo, and GPT-3.5 Turbo. Our approach involved the development of a comprehensive QA dataset, using a combination of standard evaluation metrics (Accuracy, Precision, Recall, F1-score, and AUC-ROC) for evaluating the correspondence between generated responses and source data, and implementing the post-hoc retrieval procedures to enhance factual accuracy. Through a series of detailed analyses using various visualization techniques, we demonstrated that GPT-4 consistently performs better than the other models across multiple performance metrics, including accuracy, precision, recall, and F1 score.

Our findings highlight that while GPT-4 is highly effective at minimizing false positives, it still exhibits moderate recall for hallucinated data, suggesting that some true hallucinations may be missed. This points out the importance of continuous model evaluation and optimization, particularly in applications where factual accuracy is paramount. The use of confusion matrix heatmaps, threshold sensitivity curves, and AUC-ROC curves provided comprehensive insights into the model’s capabilities and limitations, highlighting the strengths and weaknesses in detecting and mitigating hallucination. Overall, our study contributes to the ongoing efforts to improve the reliability and trustworthiness of LLMs in real-world applications. By employing a multi-faceted approach, we were able to identify key areas for improvement and propose strategies to enhance the performance of question answering systems, thereby mitigating the risk of misinformation. The comparative testing of GPT-4 with and without post-hoc search obviously shows the better results of hallucination catching and factuality. When retrieval-based correction is turned on, we can find statistically significant improvements in F1 score and precision. These results scientifically substantiate our main argument and highlight the importance of external proof in the improvement of dependability of QA systems. Ongoing or future development will seek to maximize retrieval latency and scale up real-time correction mechanisms to interactive systems including medical decision support and academic tutoring systems.

Future work can focus on expanding the QA dataset to include a broader range of disciplines and more complex queries, developing more sophisticated hallucination detection algorithms, implementing a real-time correction mechanism, incorporating user feedback into the model training process, conducting cross-model validation studies, and investigating the ethical and societal implications of hallucinations in LLMs. These efforts will further enhance the robustness and reliability of LLMs, making them more effective tools for various domains, from health care to education and beyond.

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RANA HASSAN AJMAL is currently pursuing the Ph.D. degree with the Department of Computer Science, Government College University, Faisalabad, Pakistan. His research interests include natural language processing, deep learning/machine learning in computation, and DL on other applied challenging topics.



MUHAMMAD UMER SARWAR is an Assistant Professor with the Department of Computer Science, Government College University, Faisalabad. His research interests include database systems, text, and data mining.



MUHAMMAD KASHIF HANIF received the Ph.D. degree from Hamburg University of Technology. He is an Associate Professor with the Department of Computer Science, Government College University, Faisalabad. His research interests include big data analytics, scientific computing, and bioinformatics.



MUHAMMAD IRFAN KHAN is an Assistant Professor with the Department of Computer Science, Government College University, Faisalabad. His research interests include database systems, text, and data mining.

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